# Collision Resolution Techniques (Open Addressing & Chaining) Data Structures

> **Definition:** A **Collision** occurs in hashing when a hash function maps two distinct keys $k_1

e k_2*

---

## 1. Detailed Technical Explanation

```

COLLI

---

## 2. Technique 1: Open Addressing

All keys are stored directly inside the hash table array. If slot **h(k)** is occupied, systematic probing finds the next vacant slot.

### 1. Linear Probing

Probes slots sequentially with an offset of 1:

```

### 2. Quadratic Probing

Probes slots using a quadratic polynomial:

```

### 3. Double Hashing (Best Open Addressing Technique)

Uses two independent hash functions **h_1(k)** and **h_2(k)**:

```

---

## 3. Technique 2: Separate Chaining

Each slot in the hash table points to the head of a **Linked List** storing all colliding keys mapped to that index.

```

Hash T

### Separate Chaining vs Open Addressing Comparison:

| Feature | Separate Chaining | Open Addressing |

| :--- | :

---

## 4. Quick Recall Flow

```

Collisio
bucket